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SARDAR PATEL UNIVERSITY

Master of Science (IT) – SECOND Semester Examination

PS02EINT52: Data Science

19/04/2025 , Saturday



Time: 10:00 A.M TO 01:00 P.M

Marks: 70

Q-1 Choose the most appropriate answer for given Multiple Choice Questions. 08

1. Data represented in an email is an example of ____ type of data
a. Structured b. Semi-structured c. Unstructured d. All of given
2. _____ is the graphical representation of an information and data.
a. Machine learning b. Data visualization c. Data Discovery d. None of given
3. _____ is common tool for model planning.
a. SPSS b. WEKA c. R d. All of given
4. In _____ phase of life cycle of data science, there is a need to frame the business problem and formulate initial hypotheses (IH) to test.
a. Data Discovery b. Data Preparation c. Data Modelling d. None of given
5. The term Varacity means ____
a. Data can be any type b. How much data is reliable
b. c. Data has value d. None of given
6. Which of the following is NOT a data type in R?
a. Numeric b. Character c. Dictionary d. None of given
7. _____ is an integrated development environment (IDE) for R.
a. IDE R b. RStudio c. a and b both d. None of given
8. _____ option can be used to measure mass of an object.
a. liter b. meter c. kb d. None of given

Q-2 Answer the following question in brief. (Attempt Any SEVEN) 14

1. List out the steps to analyse unstructured data.
2. Which are three the most important characteristics, a data scientist should have?.
3. List out at least FOUR unstructured data analytic tools.
4. With at least TWO example(s) explain the word Big Data.
5. Explain briefly the concept of Block Chain technology.
6. Define: Big data Analytic
7. List out the data quality traits.
8. What is Pandas?
9. Define Terms: I).Numpy II). Scipy

Q-3 Do as directed:

(A) List out at least FOUR importance of data science in modern business and Explain any TWO in detail 06

(B) Explain briefly (Any TWO): 06

i. Data Ops ii. Machine learning iii. Internet of Things (IOT)

OR

(B) Explain briefly the life cycle of data Science. 06

Q-4 Do as directed:

(A) Write a note on types of digital data with proper example(s). 06

(B) Explain briefly the need of Big data. 06

OR

(B) Write short note on characteristics of Big data. 06

Q-5 Do as directed:

(A) Explain briefly the classification of Big data analytic techniques. 06

(B) Explain briefly at least THREE advantages of Big data analytic. 06

OR

(B) Write short note on: Importance of Big data analytics. 06

Q-6 Do as directed:

(A) What is R? Explain R data structures in detail. 06

(B) What is Matplotlib? With example(s) explain different types of plots in Matplotlib. 06

OR

(B) Explain in detail data cleaning. 06

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